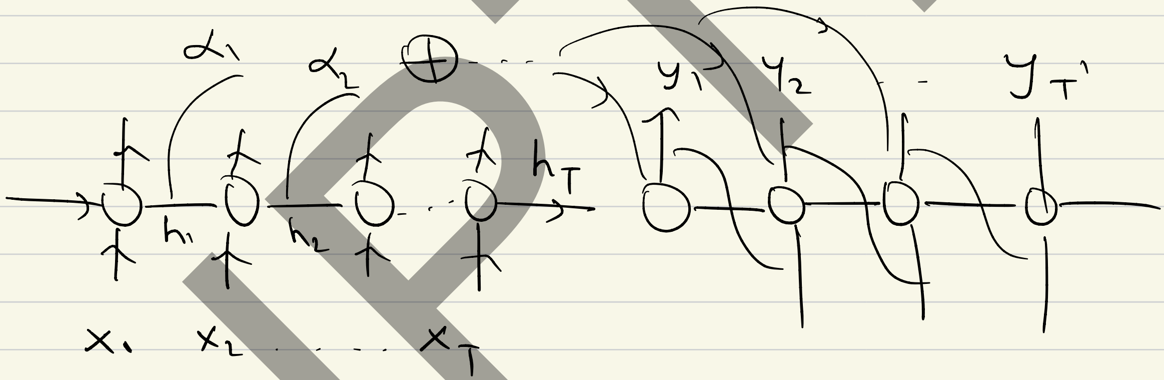
Attention Mechanism & Transformers.

$$X \xrightarrow{\phi} Z$$

Representations / Embeddings.

$$h(x) = \sigma\Big(\underbrace{W_2\sigma(W_1 x)}_{\phi(x)}\Big)$$

Attention :



$$y_t = F\left(y_{t-1},\ \sum_{t=1}^{T}\alpha_t h_t\right)$$

Given a data point

$$x = \{x_1, x_2, \ldots, x_T\}, \quad x_i \in \mathbb{R}^d \quad \text{(token)}$$

<u>Objective</u> : Learn a projection / representation that encodes the interaction b/w the tokens.

Let X be a matrix, having the data

$$X \in \mathbb{R}^{T \times d}$$

Define linear projections of X using three matrices.

$$Q = \overset{T \times d}{X} \overset{d \times d_q}{W_q}$$

where W_q, W_v & W_K are learnable.

$$K = X \overset{d \times d_k}{W_K}$$

$$V = X \overset{d \times d_v}{W_v}$$

Assume $d_k = d_v = d_K$

<u>Defining the Attention vector.</u>

Given a row of Q, denoted by $q \in \mathbb{R}^{d_v}$

compute the following to get the "attention".

i) $s_i = q k_i^T \quad \forall \; k \in K., \; i = 1 \dots T$

Let $\underset{1\times d_v}{q} \underset{d_v \times T}{K^T}$, $S = qK^T$, a T-length vector, quantifying the similarities b/w q & all rows of K.

$$\hat{s} = \frac{s}{\sqrt{d_v}}$$

$$\alpha = \text{softmax}(\hat{s})$$

α represents the scaled similarities b/w q & all the elements of K.

Define attention vector A_q for q as

$$A_q = \sum_{i=1}^{T} \alpha_i v_i$$

q : query $\quad K$: Key $\quad V$: value.

$$\text{Attention}(q, K, V) = \left[\text{softmax}\left(\frac{qK^T}{\sqrt{d_v}}\right)\right] V$$

$$\underset{T\times d}{X} \xrightarrow{\text{Attention}} \underset{T\times d_v}{Z}$$

$$\phi(X)$$

Multi-head Attention

Learn multiple attention projections on X, call them $z_1, z_2 \ldots z_m$

where $z_j : \text{Attn}(X)$

Define

$$\underset{T \times d_v}{\text{MHA}(X)} = \underset{T \times (m \times d_v)}{[z_1, z_2 \ldots z_m]} W_2.$$

$$h_\theta(x) : X \to Y$$

regularize(θ).

$$h_\theta(x) = W_2 \sigma(W_1 x) \qquad \theta = \{W_1, W_2\}$$

$$= W_2 z.$$

$$z = \sigma(W_1 x)$$

$h_\theta(x)$ can be viewed as a fn. of θ & z

Normalizations are used to regulanze z.

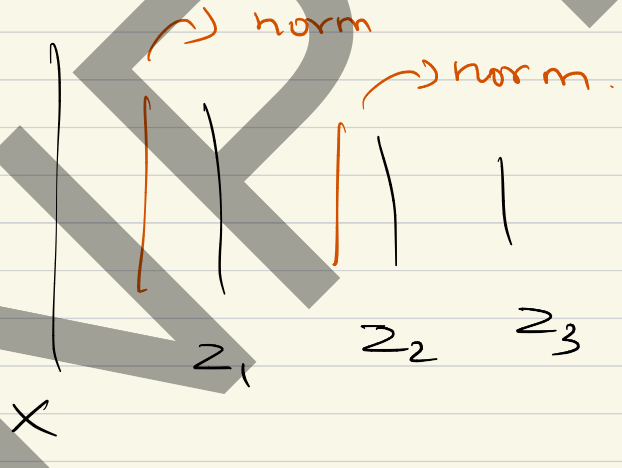
i) Batch normalization

ii) layer normilization.

$$\hat{z} \leftarrow \left(\frac{z-\mu_z}{\sigma_z}\right)\gamma + \beta$$

$$\mu_z = \frac{1}{B}\sum z_i$$

$$\sigma_z = \mathrm{var}(z_i)$$



Transformer Architecture.